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IFPRI Discussion Paper 01683

November 2017

The Central Position of Agriculture within the 2030 Agenda for Sustainable Development

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INTERNATIONAL FOOD POLICY RESEARCH INSTITUTE

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ABSTRACT

This paper provides a helpful framing to understand both why and how policy attention and investments should be channeled through agriculture and agrifood systems as key vehicles for achieving the Sustainable Development Goals (SDGs). It illustrates the ways in which agriculture, particularly within the context of food value chains, is uniquely positioned to holistically address the social, economic, and environmental dimensions of sustainable development because of its existing reach and future potential. In this paper's examination of the multiplicity of entry points the sector offers for fostering inclusive and sustainable economic growth, reversing harmful environmental trends, and enhancing the resilience of the poorest and most vulnerable populations, it traces some of the most potent pathways for agricultural policies and interventions to accelerate development outcomes across all country contexts.

Keywords: Agenda 2030, SDGs, agriculture, poverty, food system, food security, fighting famine and hunger, sustainable development, environment, economic development, social development, gender, inequality, nutrition, resilience, climate change

1. INTRODUCTION

Background

The progress achieved under the Millennium Development Goals (MDGs), described as the “greatest anti-poverty push in history” by former UN Secretary General Ban Ki-moon, underscored the value of an aspirational global agenda grounded in concrete and measurable targets for accelerating poverty reduction. As we have entered a new era in development cooperation, marked by the adoption of the 2030 Agenda for Sustainable Development in 2015, there has been a revitalization of global partnership for development. Not merely a repackaging of their predecessor, the Sustainable Development Goals (SDGs) have translated the wealth of lessons and experiences from the previous era into an unprecedented new agenda that is supremely ambitious in its vision.

Building on the unfinished business of the MDGs, the SDGs represent a transformative framework based on sustainability in all its dimensions, as well as a plan of action for people, the planet, prosperity, and peace. Most critically, the new framework reflects an indivisible and integrated set of goals that represent a declaration of interdependence (Levy-Abegnoli 2016). By design, they underline the imperative of responding to every development goal with equal urgency and attention, in recognition of the deep interlinkages and multidirectional relationships that tie them all together.

No more evident is this indivisibility than in the relationship between SDG 2 and all the other goals. Highlighting the interconnectedness among food security, nutrition, rural transformation, and sustainable agriculture, SDG 2 aims to end hunger and eradicate all forms of malnutrition by 2030. From strengthening the sustainability of food systems for more efficient and environmentally sound production and consumption to maintaining genetic diversity and empowering smallholder farmers, the articulation of the targets under SDG 2 recognize “that we can no longer look at food, livelihoods and the management of natural resources separately” (FAO 2016a, 2016b). Whereas the MDGs had grouped the reduction of poverty and hunger as one goal, their presentation as interrelated yet distinct goals under the SDGs allows for a heightened emphasis on the multidimensionality of hunger and malnutrition and a more nuanced and

expansive treatment of agriculture and food systems. Within this framing, agriculture emerges as central not only to the achievement of zero hunger and malnutrition but also to the entire agenda.

As both a direct and indirect pathway to the social, economic, and environmental dimensions of sustainable development, agriculture, “more than any other sector, ... is the common thread which holds the 17 SDGs together” (Farming First 2017). These interlinkages are found, in part, through the explicit reference to agriculture across multiple SDG targets and indicators, such as in the SDG 5 target relating to women’s improved access to agricultural land or the SDG 14 targets pertaining to sustainable fishing and small-scale fishers. Understood in terms of the full range of farming and livelihood activities tied not only to crop production but also to livestock, forestry, fisheries, and aquaculture, agriculture likewise appears directly implicated, even when not explicitly referenced, in the SDGs relating to the sustainable use of natural resources, ecosystem management, climate change adaptation, and environmental protection more broadly (FAO 2016b).

Apart from the sector’s importance at the production level, the conceptualization of agriculture as one point along a larger value chain, particularly in the context of food systems, places a critical spotlight on the manifold entry points and actors implicated by the flow of agricultural goods and services along the entire chain, from producers to consumers. The SDG 2 targets help reinforce this system-level perspective of agrifood chains, including through its focus on issues of agricultural trade and markets that shape the surrounding institutional and policy environment. Through this system approach, the interlinkages between agriculture-related investments and the other SDGs appear more pronounced—for example, as seen through relatively direct relationships, such as the importance of efficient value chains in reducing food loss and waste, as outlined by SDG 12 on sustainable consumption and production, and through a wide range of indirect connections.

In particular, the rural–urban linkages embodied by agrifood value chains speak to the interrelationship between rural and urban development, including the ways in which food systems shape diets in both cities and rural settings or the ways in which rural transformation spurred by agriculture can contribute to growth in nonagricultural sectors and urban contexts. Interventions tied to agrifood systems

are thus inextricably part of objectives contained in SDG 9, relating to resilient infrastructure and inclusive industrialization, and SDG 11, concerned with sustainable urbanization. In this way, in part through a framing focused on agrifood systems, Agenda 2030 helps expand the conceptualization of agriculture within development, maintaining its fundamental relationship with poverty reduction, while also extending a spotlight to both its direct and indirect connections to other SDGs affecting people, the planet, peace, and prosperity. Although this paper only focuses on some of these connections, other works have further examined the multidirectional relationships that exist between food and agriculture and the other SDGs. One such outline may be found in the FAO publication *Food and Agriculture: Key to Achieving the 2030 Agenda for Sustainable Development* (2016b).

Narrowing the Focus to Widen the Impact

Because the indivisibility of the SDGs necessitates a simultaneous focus on all goals to sustainably achieve any one goal, the reality of limited resources and capacity requires governments to prioritize initial investments into key policy and program areas with the most potential to accelerate progress across multiple SDGs (Omilola 2017). Given its universal relevance, agriculture represents a strategic entry point for accelerating development priorities both locally and globally, whether through the development of the agriculture sector and policy reform at the national level, investments in agriculture systems through international cooperation, or global efforts to render agricultural markets and food production systems more just and sustainable. From helping countries achieve growth and poverty reduction to enabling them to reduce their environmental footprint, this potential for agriculture to greatly advance SDG implementation stands just as relevant to the least-developed countries as to the high-income ones. Despite this reality, however, levels of investment in inclusive and sustainable agriculture remain inadequate. The Agriculture Orientation Index, for example, shows that agriculture's share of government expenditures relative to its share of gross domestic product (GDP) "fell from 0.38 in 2001 to 0.24 in 2013 and 0.21 in 2015 worldwide" (HLPF 2017a). Within Africa, total public spending on agriculture as a share of public spending fell far short of the Comprehensive Africa Agriculture Development

Programme's target of 10 percent in 2014, revealing a divergence between commitments and action (IFPRI 2016).

Taking the SDG framework as a conceptual basis, this paper develops several arguments to make the case for the centrality of agriculture in achieving the vision embodied by Agenda 2030. As earlier suggested, the discussion of agriculture's role cannot be separated from that of agrifood systems. Given its limited scope, this paper primarily focuses on the significance of agricultural production, most notably in its discussions of economic and environmental impact. Yet, it widens analyses throughout to also consider the inherent interactions between agriculture and other areas of development in relation to the agrifood value chain in its entirety, including recognizing all the actors and processes relating not only to agricultural inputs and production but also to storage, processing, distribution, transport, retail, and consumption (IFPRI 2017).

Although the sustainability and inclusiveness of production activities are crucial, they do not function separately from the rest of the chain. The potential for agriculture to contribute to enhanced economic opportunities for smallholder farmers, for example, depends not only on investments in inputs or on capacity building for increased productivity but also on smallholders' ability to access markets, just as consumer preferences and dietary behaviors influence smallholders' production choices and capacities. As will be further illustrated, we similarly cannot speak of food loss and waste, agricultural interventions for enhanced nutritional outcomes, or rural–urban linkages without considering the various multidirectional interactions that appear at each stage of agrifood systems.

Bearing this connection to systems in mind, the importance of agriculture, as both an urgent necessity and a significant opportunity for development, lies, on the one hand, in its unique capacity to holistically address the social, economic, and environmental dimensions of the SDGs, as the “fundamental connection between people and the planet, and the path to inclusive and economic growth” (FAO 2016b). On the other hand, agriculture's centrality to the global agenda is evident in the scope of its existing and potential negative toll on other areas of development if immediate action is not taken, particularly with respect to environmental impact. This dual analytical perspective—underscoring

relevance not only through potential entry points but also in recognition of existing harm—provides a helpful framing to understand both why and how policy attention and investments should be channeled through agriculture and food systems.

The first section of this paper illustrates the “why,” revealing the scope of agriculture’s impact on people and the planet through global trends and projections and demonstrating the extent to which the sector can address major development challenges based on its existing reach. The sections that follow deconstruct the critical pathways and opportunities for agriculture and agrifood systems to contribute simultaneously to the interconnected social, economic, and environmental drivers of development, both at the macro and the micro levels. Among these opportunities includes the capacity for action on agriculture to drive forward robust local, national, and global multistakeholder, multisector partnerships by embodying natural entry points for engagement across multiple sectors, as well as with the most marginalized and vulnerable segments of society.

2. THE URGENCY FOR ACTION IN AGRICULTURE: FUTURE POTENTIAL BASED ON CURRENT TRENDS

Although few would argue against the important role of agriculture and agrifood systems in Agenda 2030, the extent of its value to both local and global sustainable development plans warrants further attention. An examination of agriculture's scope of influence, including through the actors it implicates along all points of the value chain, suggests expansive opportunities for the acceleration and multiplication of development outcomes. Several characteristics underline the uniquely powerful position that agriculture holds relative to other sectors and development pathways: the massive scope of its environmental footprint, its connection to peace and stability, and its direct link to the livelihoods of most of the world's poorest populations.

While these broad relational categories for framing the potency of agriculture are not exhaustive, they are particularly apt for describing its crosscutting importance to the global development agenda in considering their common underlying connection to resilience. As a “useful operational framework for reducing the multiplicity of risks faced by people and communities, now and in the future,” the concept of resilience stands central to the achievement of the SDGs and the principal of leaving no one behind (Bahadur et al. 2015). Having the greatest impact on the poorest and most vulnerable communities, the increasing frequency and severity of shocks of all forms continue to undermine development efforts, ranging from conflict and displacement to food price volatility, climate change, and natural disasters. Most visible through the connection of agriculture to livelihoods, the environment, and human security, multiple opportunities exist to advance the resilience agenda by helping people, communities, countries and global institutions prevent and recover from shocks (Fan, Pandya-Lorch, and Yosef 2014).

Many of the deepest drivers of vulnerability to the often-overlapping threats of social, political, economic, and environmental shocks can be addressed, in part, through agricultural or agrifood pathways. As the following sections explore, these pathways can include enhancing the livelihoods of smallholder farmers and the poorest populations, which boosts their capacity to withstand economic shocks;

leveraging interventions in agriculture to both mitigate and increase adaptation to climate change; and increasing rural and urban populations' nutrition security (Fan, Pandya-Lorch, and Yosef 2014).

Environmental Impacts

The environmental toll of agricultural production alone illustrates the urgency of shifting to more sustainable systems and the significant outcomes that would result from better investments. It is estimated that “20–25 percent of total greenhouse gas emissions are directly caused by agriculture, including deforestation” and other types of land use (UNDP and UNRISD 2017). This estimate has nearly doubled over the past fifty years; in the absence of immediate measures to reverse these trends, emissions could rise by another 30 percent by 2050 (Michalopoulos 2016).

Beyond its role in climate change, agriculture is equally culpable in the destruction of natural resources. Considering that agriculture uses almost one-half of the planet's vegetated land as well as much of the world's available fresh water, with about 85 percent of global water use for irrigation, it is apparent that the future of our terrestrial and water ecosystems depends upon the future of agriculture (WRI 2013). Already, 33 percent of land in the world is highly degraded, with agriculture contributing to the loss of approximately 20,000–50,000 kilometers of potentially productive lands each year and agricultural expansion constituting by far the leading cause of tropical deforestation (HLPF 2017a; Giovannucci et. al. 2012). Meanwhile, roughly two-thirds of the global population live in areas with severe water shortages for at least one month of the year. It is estimated that the share of people living under conditions of severe water shortage will increase to 52 percent by 2050 if water resources are not managed efficiently (HLPF 2017a; IFPRI 2016). These impacts are not inevitable. Rather, they are the result of unsustainable consumption and production behaviors, as seen by the fact that a third of all food produced is lost or wasted and at least 15–35 percent of the global water use for agricultural irrigation is considered unsustainable (IFPRI 2016).

Compounded by a range of global trends, including extreme population growth, the increasing frequency of natural disasters, the widespread occurrence of conflict, and shifting consumption patterns

tied to economic growth, the already dire picture for the environment appears even more bleak for the future if the current trajectory of unsustainable agricultural practices continues. The United Nations projects that the “world’s food producers will need to produce 70 percent more food by 2050 to feed a projected world population of 9.6 billion” (IFPRI 2016). At the same time, the environmental costs of agriculture function as a vicious cycle, putting further pressure on production capacities to meet increasing consumption demands. It is projected that land degradation, for instance, may reduce global food production by up to 12 percent in the next two decades, which, in turn, will further constrain food access for the poorest through an increase in world food prices of up to 30 percent (UNDP and UNRISD 2017). Shifting to more sustainable agricultural approaches is thus the only way to feed the rapidly growing global population, while also ensuring that intensified production does not further deplete the earth’s limited resources (Giovannucci et al. 2012).

Relationship among Agriculture, Peace, and Stability

Beyond the implications for the planet’s health and security, failure to adequately invest in sustainable agriculture would likewise undermine any efforts to secure the health and stability of societies in every sense. The increasing incidence of natural disasters and erratic weather patterns induced by climate change results in the destruction of livelihoods and the displacement of populations. These sources of insecurity exacerbate poverty and hunger, which function as underlying drivers of civil unrest and conflict, along with heightened tensions over increasingly scarce natural resources. All of this, in turn, weakens the capacity of institutions and the establishment of good governance that is necessary for long-term development. At the same time, conflict and the displacement it causes destroy the physical and human capital that allows for a healthy food system and resilient communities (IFPRI 2017). With the number and intensity of conflicts, crises, and natural disasters on the rise, as evidenced now by the 20 million people in four countries (South Sudan, northeastern Nigeria, Somalia, and Yemen) facing severe food insecurity and impending famine in what has become the largest humanitarian crisis since World

War II, the relationship between agriculture and peace and stability appears all the more noteworthy (FAO 2017).

While the role of food insecurity has received considerable attention within global discourses on the humanitarian–development–peace and security nexus, to the extent that some claim it may become the leading cause of conflict in this century, it is necessary to recognize that food insecurity constitutes just one among many drivers tied to agriculture (ICSU 2017). In addition to connections through food emergencies, links of agriculture and food systems to poverty alleviation, governance, and resource scarcity or abundance more broadly underscore their potential to influence a range of factors that instigate, perpetuate, or prevent conflict, including post-conflict reconstruction efforts (Taeb, 2004).

Connection to the Livelihoods of the Poor and Vulnerable

Just as agriculture plays a decisive role in the fulfillment of the SDGs’ environmental targets, without which the sustained peace and prosperity of societies will be impossible to achieve, the scale of its existing and potential influence is likewise demonstrated through its connection to livelihoods. This pathway particularly underscores the unique capacity for the sector to operationalize one of the most challenging yet fundamental elements for the implementation of the SDGs: the commitment to leave no one behind. While similar arguments could be made based on other vehicles for human development encompassed by agriculture, the most obvious being its status as the source of food and nutrition, a focus on the relationship between poverty and agriculture through the lens of livelihoods is particularly revelatory of agriculture’s potential to empower those who are the hardest to reach.

The most telling indicator for this argument lies in the average profile of those who are employed by or depend upon agriculture for their means of living. Poverty today is predominantly rural: some 80 percent of the extreme poor live in rural areas, and 65 percent of the extreme poor work in agriculture (IFAD 2016). Among this population, most are smallholder farmers who lack access to the level of resources and markets enjoyed by larger producers; thus, they fail to achieve the most basic level of subsistence. Small farms support the livelihoods of up to 2.5 billion people worldwide, and account for up

to 80 percent of the food produced in Asia and Africa south of the Sahara (IFAD 2016). In simply considering the composition of the populations most employed by the sector, it is clear that poverty eradication policies and strategies cannot succeed without a focus on agriculture, including its role in contributing to inclusive rural transformation.

While agriculture provides an important engine for promoting the economic inclusion of those who have been left furthest behind by development, given these linkages to employment, its potency extends beyond addressing income-related factors of inequalities and exclusion. Again, this may be seen in the composition of those who are most dependent upon agriculture, which reveals a web of multiple intersecting vulnerabilities grounded in political, economic, identity, and geographic dimensions of inequality, producing pockets of marginalization and exclusion among the poorest of the poor. Identity characteristics such as age and gender, for example, have a profound influence on farmers' production abilities and income-earning potential, while geographic factors such as drought or climate shocks exacerbate these identity-based or political and economic inequalities, most severely impacting those with the least resources and adaptive capacities.

Inasmuch as concerted investments and policies to address the specific constraints of smallholder farmers as a target group will have significant implications for the achievement of national social and economic development priorities, further opportunities for reducing inequalities exist in considering the range of disadvantaged groups within this larger category. No doubt, gender inequalities represent the most pervasive form of inequality, with sociocultural norms and policy environments dictating access to resources and preventing women farmers from achieving the same productive capacity and income-earning potential as men, despite making up 43 percent of the world's producers (IFPRI 2016).

Another predominant driver of social exclusion and economic disadvantage is aging, which tends to affect the productivity of rural dwellers. It is, therefore, important to strengthen the well-being of older populations through agriculture (FAO 2017). Equally striking is the fact that, globally, "60 percent of all child labourers in the 5–17 age group are engaged in agriculture," with children often taken out of school to work on subsistence farms or to generate income for the family; this highlights the potential to improve

education outcomes and break cycles of poverty through interventions in agriculture (ICSU 2017).

Enhancing agriculture livelihoods with a focus on equity can thus go a long way not only in contributing to pro-poor growth but also in eliminating many of the structural drivers that continue to hold back certain groups from achieving their full socioeconomic potential.

3. DELIVERING ON THE ECONOMIC, ENVIRONMENTAL, AND SOCIAL DEVELOPMENT GOALS

As the preceding section illustrated, a focus on scope and reach of influence, both positive and negative, underscores the central position already held by agriculture and agrifood systems in the context of the SDGs. Most notably, the impact of these processes on people and the planet, as is particularly visible in the context of environmental sustainability and the development of rural livelihoods, underlines the capacity for agriculture to operationalize two core aspects for successful implementation of Agenda 2030: the principle of universality and the call to leave no one behind.

The current context of poverty, environmental degradation, hunger, and conflict, as the facts and trends reveal, ultimately speaks to the direct and profound implications that investments in agriculture and food systems will have on the immediate survival of millions of people, the structural transformation and long-term development of nations, and the extent to which development gains will benefit all countries and people equitably. This analysis of the extensive costs borne out of unsustainable agricultural practices and inadequate investments already provides a compelling case for action and impetus for fostering political will and partnerships; a focus on the multitude of development pathways offered by agriculture will further complement this perspective to produce a more complete picture of its potential contributions to Agenda 2030.

The remainder of this section explores the most potent entry points by examining agriculture's necessary role in accelerating progress across the social, economic, and environmental dimensions of the SDGs, while also serving as a natural bridge between each interdependent area.

Economic Pathways

Although the number of people living in extreme poverty dropped by more than half between 1990 and 2015, from 1.9 billion to 836.0 million, lifting the nearly 800 million people who still live on less than \$1.90 a day out of poverty remains the greatest global challenge and an indispensable requirement for sustainable development (UNDP and World Bank, 2016). In its commitment to completely eradicate all

forms of poverty by 2030, placing this goal at the heart of the SDGs, the international community has signaled its resolve to finish what the MDGs had started. Although political will and investments, along with rapid industrialization in several populous countries, were sufficient to bring about the drastic drop in poverty over the past decades, when many of the poor lived just below the poverty line, the same broad-based policy reforms and approaches will not yield the same outcomes for those who continue to live in extreme poverty. Many of these people reside in remote communities, largely disconnected from society, and belong to marginalized groups facing multiple, compounding sources of social and economic discrimination (UNDP 2016a).

Likewise, inequalities between countries reveal how the progress achieved under the MDGs was far from even, leaving the majority (80 percent) of the world's extreme poor concentrated in only two regions—Africa south of the Sahara and South Asia (UNDP 2017). In fact, Africa south of the Sahara saw the number of people living in extreme poverty increase from 290 million in 1990 to an expected 403 million in 2015, in part due to rapid population growth (UNDP and World Bank 2016). These dynamics have led to the widespread recognition, as embodied by the focus on equity across the SDG targets, that only inclusive and equitable growth that places the needs of the poor and marginalized at the center of development priorities will enable countries to eradicate multidimensional poverty.

Economic Growth That Benefits the Poor

Accounting for one-fourth of GDP in developing countries, agriculture constitutes a powerful engine for accelerating economic growth, particularly as a vehicle for rural transformation, employment creation in both rural and urban areas, and job diversification (FAO 2016b). Yet, while the growth of national economies spurred by agriculture forms a crucial basis for human development, it does not guarantee that the benefits will trickle down equally to the entire population. In this regard, the multifaceted potential for agriculture to contribute to pro-poor growth and poverty reduction, while also boosting GDP growth, highlights its centrality to Agenda 2030 and its necessary role for the fulfillment of SDG 1 of eradicating poverty in all its forms everywhere by 2030.

There is compelling evidence to suggest that agricultural growth is, on average, at least twice as effective in reducing poverty as growth generated in non-agricultural sectors, five times more effective than other sectors in resource-poor low-income countries (excluding Africa south of the Sahara), and eleven times more effective than other sectors in Africa south of the Sahara (FAO and IFAD 2013). This effect is produced through the combination of direct and indirect economic development pathways offered by the sector—on the one hand, through the direct link between agriculture and rural livelihoods in reducing poverty directly, by raising farm incomes, and, on the other hand, through the indirect effects of employment generation outside of agriculture and the reduction of food prices for poor consumers, particularly through the increased productivity of nontradable staple foods (World Bank 2008). An emphasis on pro-poor growth strategies multiplies these direct and indirect outcomes by ensuring that those who would ordinarily not be able to participate or benefit in market integration and investment in agriculture are able to do so. These strategies will stimulate the economic sectors where the poor work, improve economic opportunities of the poorest groups, and address the root causes of migration (FAO 2017).

Structural Transformation

Whereas the direct effects on rural livelihoods and productivity can help boost income-generating opportunities for the most vulnerable in the immediate term, the capacity for agriculture to facilitate the structural transformation of economies and, subsequently, enhance the income-generating capabilities and employment opportunities available to the poor is most critical for the long-term sustainability of its poverty-reducing effects. As both a cause and an effect of economic growth, structural transformation constitutes a change in the economies of developing countries from low productivity agriculture and informality to high productivity sectors, including manufacturing, industry and modern services. Evidence from Asia shows that development of agriculture and increase of agricultural productivity should play a larger role in fast-tracking structural transformation, in view of closing the rural–urban divide. In this respect, agriculture’s linkages to the rural nonfarm economy, manufacturing and services, job creation,

and diversification in other sectors represent fundamental entry points for maximizing its potential benefits to the population at large, positioning agriculture as a bridge between rural and urban development. For example, investments in sustainable agriculture, the empowerment of smallholders, and enhanced productivity can yield multiplier effects on other industries (IFAD 2016). Such pathways underscore the interdependent relationship among agricultural growth, rural transformation, and the larger structural transformation of economies, creating a virtuous cycle that feeds back to the advancement of rural development, contributing to increases in productivity, which help improve rural livelihoods and promote inclusive growth.

Through investments that capitalize on these intrinsic pathways between rural and urban inclusive growth and that promote their economic integration, agriculture can also help address rural–urban disparities and encourage the retention of skilled people, including youth, in rural areas (UN-DESA 2009). Considering that more than 200 million people around the world are unemployed, including 74 million young people (those age 15–24), the employment creation potential latent in agriculture is a fundamental asset for realization of the SDGs (UNDP 2017). Within rural areas, this potential job creation extends beyond farming, agro-processing, and rural industry to include building infrastructure and sustainably managing resources, waste, and residues (UN-DESA 2009).

On a broader level, investments in rural infrastructure and in the intermediate towns between cities and rural areas can also help build linkages and stimulate economic activities, which can benefit both smallholders and those in cities (IFPRI 2017). In addition, in the face of economic shocks and crises, the agriculture sector and rural livelihoods may serve as a buffer for those in urban settings by creating more temporary jobs for rising urban population in cities during financial and economic turmoil (ICSU 2017).

Amid contexts of increasing urbanization, most notably in developing countries, with approximately 66 percent of the population expected to live in urban areas by 2050, these rural–urban linkages embodied by agriculture and food systems are critical for the SDGs (IFPRI 2017). Leveraging agriculture and agrifood systems to strengthen the physical, economic, social, and political linkages between cities and rural areas can serve as opportunities for producers, especially rural smallholders, to

improve their livelihoods, while also contributing to improved nutrition among poor and vulnerable communities by rendering food systems both more effective and more inclusive (IFPRI 2017). Some of these connections may be naturally propelled by processes of urbanization; for instance, the increase in urban demand for more and better food enables smallholders to increase and diversify their food production in rural areas (IFPRI 2017). Yet, urbanization could equally pose challenges for agricultural livelihoods by adding stress to agricultural systems if policies do not harness the synergies between rural and urban areas and ensure strong value chains (IFPRI 2017).

Part of agriculture's role in responding to evolving development trends in urbanization, population growth, migration, and shifted dynamics for production and consumption will depend on its contributions to employment in other sectors and its ability to bring rural people into the economic mainstream (IFAD 2016). With proper investments, policies, and institutional support, growth in agriculture and its enhanced integration with other sectors through markets should drive a development trajectory in which small-scale farmers increasingly integrate into commercial food systems, allowing for the gradual diversification of employment and, for many, the option to exit the agricultural sector entirely (FAO 2017). This process, which is characterized by agricultural transformation and, eventually, broad-based structural transformation, underlines the ultimate role for development of the sector not as an end in itself but as a means to enhance the range of opportunities for the poor across sectors and geographic contexts.

Closing Gaps

It can be deduced from almost all countries' decrease in the relative contribution of agriculture to their GDPs over the past 50 years that this scenario has played out to some extent around the world due to the intrinsic pathways described earlier. However, the impact of agriculture on poverty reduction within countries has nonetheless varied. Ultimately, agriculture represents an opportunity—and, for many countries in the early stages of development, a necessary condition—for economic growth and poverty reduction. As a telling indication of this relationship, a recent IFAD review of country data from the 1990s to the 2010s found that no country has succeeded in rural poverty reduction without focused

policies to promote rural transformation and enhance structural transformation, along with investments in agriculture (IFAD 2016).

Within several Asian countries in the initial stages of development, rapid increases in agricultural productivity played a larger role in fast-tracking structural transformation by taking into consideration linkages with other sectors of the economy, particularly the industrial and manufacturing sectors (FAO 2006). This pattern illustrates how agricultural development catalyzed economic development in Asia. This correlation is further revealed by the fact that parts of the world where there are high and sustained levels of investments in agriculture represent success stories of poverty and hunger reduction today (HLPF 2017a). In China, for example, rapid growth in agriculture, due to market liberalization, diffusion of technology and the acceleration of industrialization, was responsible for the rapid decline in rural poverty from 53 percent in 1981 to 8 percent in 2001 (World Bank 2008).

The effect on poverty, however, strongly depends upon the extent of inequality in a country, as reinforced by an IFAD study that observed that agriculture only reduced poverty among the poorest of the poor in contexts where income inequality was not excessive (Christiaensen, Demery, and Kuhl 2010). The Green Revolution, which was so successful in promoting economic growth in south Asia, helps illuminate these dynamics. Characterized by high rates of investment in technology, infrastructure, market development, policy support, research and extension services, the Green Revolution reduced poverty throughout the region and averted hunger for millions (Pingali 2012). Through huge boosts to productivity, the region saw remarkable growth in GDP, “averaging an annual 4.5 per cent in 1980–2000 and 4.4 per cent in 2000–2013, compared with 2.6 per cent and 2.0 per cent in the rest of world over these periods” (IFAD 2016). Nevertheless, these massive gains failed to solve the poverty, food security, and nutrition problems that continue to face poor societies, in part due to a limited equity focus within the adopted policies for agricultural intensification (Pingali 2012). As the experience of India and other Asian countries has shown, macroeconomic and agricultural policies are fundamental in shaping how the benefits of overall growth are distributed (FAO 2006). Investments in agricultural growth must be coupled with policies that facilitate equitable land distribution, promote better access to assets and

markets, increase technology transfers to women, support smallholder farmer growth, and protect the management of natural resources, among other issues (Pingali 2012).

Environmental Pathways

While agriculture may accelerate countries' poverty reduction efforts, this progress will be for naught if investments in increased productivity are not coupled with those to raise its sustainability. Bringing the poverty-environment nexus to the forefront of the development agenda for all countries, Agenda 2030 emphasizes the fundamentality of sustainable management of ecosystems and natural resources (SDGs 6, 14, 15), sustainable consumption and production patterns (SDG 12) and urgent action on climate change (SDG 13) as integral for the achievement of all development objectives (UNDP and UNRISD 2017). Because the planet's natural resources play a role in food production, livelihoods, and economic systems, they underpin the delivery of all SDGs (Declercq 2017). Agriculture is the most significant user of the earth's scarce resources; the primary cause of land use change, land degradation, and desertification; and the sector that accounts for nearly a fifth of global greenhouse gas emissions; thus, it sits at the heart of the environmental priorities outlined by the SDGs (ICSU 2017).

Just as agricultural production influences soil and water quality, land use, and ecosystem health and functioning, the sustainable management of the earth's resources, reversal of environmental degradation, and the mitigation of climate change will determine agriculture's capacity to meet the growing, changing demand for food and to support the livelihoods, health, and well-being of all populations (ICSU 2017; FAO 2017). As one of the most powerful examples, climate change will greatly undermine poverty reduction efforts, largely through its negative impacts on agricultural-sector incomes, potentially contributing to 5 percent of global crop yield losses by 2030 and potentially resulting in an increase in the number of people living in extreme poverty by 122 million by 2030 (UNDP and UNRISD 2017).

Fortunately, the consequences of inaction are just as great as the potential opportunities from increased investments in agriculture that balance the needs of both planet and people. Just as the adoption of the 2030 Agenda reinforced a paradigm shift in approaches to development that consider all objectives

as multidimensional and intricately linked to social, economic, and environmental factors, approaches to agriculture must reflect this interconnectivity and respond to the emerging trends and global challenges—from the rapidly rising population growth to the devastating consequences of climate change to the emergence of dietary habits that threaten both the health of people and the ecosystem.

This rethinking of agriculture should ultimately move away from a focus on more production to one on better production. Doing so requires creativity and innovation in efficient management of key resources such as land, water, energy and inputs for food production (Giovannucci et al. 2012). Most notably, through financial commitments and policy interventions geared toward mitigating the environmental impact of intensified production, coupled with efforts to leverage agriculture to build the resilience of communities to climate change and other environmental shocks, countries can alter the position of agriculture from being part of the problem to part of the solution.

Mitigation Opportunities

Creating an efficient, sustainable system capable of producing more food with the fewest possible resources stands out as one of the most urgent and daunting challenges for realizing the SDGs (ICSU 2017). In the broadest sense, the concept of sustainability at the heart of the global development agenda pushes governments and development actors to widen their vision, moving away from myopic approaches of the past that failed to consider the wider or future impacts of responding to a certain need (Sustainable Development Commission 2011). Applied to agricultural systems, it is important to balance environmental, social and economic considerations by building systems capable of meeting current and future demand for food without compromising the availability of scarce natural resources (IFPRI 2016).

As is visible in the extensive damage to land, water, and the atmosphere directly caused by agriculture in its present state, much of the focus has been on intensifying production at the expense of the environment. The costs to the environment will become exponentially worse, and harder to reverse, if current patterns of consumption and production, particularly through agriculture, remain unaltered, with a projection that by 2050 the global use of natural resources will be four times the amounts of 2010 (UNDP

and UNRISD 2017). At the same time, agriculture has been projected to consume 70 percent of the total permissible ‘budget’ of greenhouse gas emissions (WRI 2013). Considering the need to close the 70 percent gap between the amount of food currently produced and what will be needed to feed the projected nine billion people on the planet by 2050, how countries respond to this urgent challenge will determine not only the speed by which the world is able to reach its environmental goals but also its ability to meet them at all (WRI 2013).

Despite inadequate investments thus far to secure this fate, the availability of research and technologies for the development of low-carbon, climate-resilient agriculture processes, combined with the wealth of evidence for their projected influence on social, economic, and environmental factors of development, suggests that a reversal of current trends is completely feasible, dependent only upon sustained investments in their application (UNDP and UNRISD 2017). One avenue includes measures to limit the expansion of land devoted to agriculture, which, since the 1960s, has grown globally by more than 10 million hectares per year and which will require an estimated additional 100 million hectares between now and 2050 to meet the rising consumption needs (WRI 2013; FAO 2017). Intensified production, when implemented sustainably, can help reduce the need to expand into new land, prevent deforestation, and limit greenhouse gas emissions, while also helping promote the health of the soil and encourage sustainable land management (UNDP and UNRISD 2017; IFPRI 2016). Researchers have identified a number of agricultural technologies that can facilitate sustainable intensification, thus reducing trade-offs and optimizing synergies between increased production and environmental health. Examples include nitrogen-use efficiency, heat- and drought-tolerant crop varieties, precision agriculture, and drip irrigation (IFPRI 2016).

In addition to technological options and research to promote farming “smarter,” a range of existing policy options can influence behaviors and combat unsustainable approaches to farming. These include “a moratorium on further expansion of agricultural areas into tropical forests or peatlands, a tax on highly emitting livestock production systems,” and policies to promote the adoption of no-till agriculture and expansion into low-carbon, degraded lands (ICSU 2017). In addition to mitigating environmental damage, the implications of such shifts would be significant for food security, as “restoring

just 12 percent of the world’s current degraded agricultural land could potentially feed 200 million people by 2030” (UNDP and UNRISD 2017).

Another critical dimension of sustainability rests in enhancing the efficiency of agrifood systems, particularly through reductions in food loss and waste. The environmental costs of the estimated 30–40 percent of food that is lost or wasted globally along all steps of the food cycle—with 24 percent occurring at production, 24 percent during handling and storage, and 35 percent at consumption—are immense when we consider the amount of land and other resources channeled into the production of food that will never be eaten (IFPRI 2016; WRI 2013). Although much will depend on shifting behaviors on the consumer’s end, investments on the production side will also be critical. Many of these potential interventions will likewise help boost the capacities of low-income, smallholder farmers, particularly in developing countries, where more losses occur during production and storage. For instance, policies geared toward small-scale producers to improve the availability of agriculture extension services, market access, and harvesting techniques or to enhance access to low-cost handling and storage technologies can help cut down losses at the production end of the value chain (WRI 2013). While helping to save the planet, cutting food losses and waste will also have an immediate and expansive effect on food availability, with the potential to reduce the food gap by approximately 20 percent by 2050 if the losses and waste are cut in half (WRI 2013).

As another form of waste, the amount of water lost to unsustainable, inefficient agricultural practices presents a major cause for concern, considering that more than 2 billion people live in countries with excess water stress (UNESCO 2017). In addition to the adoption of technologies proven to reduce water loss, such as lining irrigation canals, the use of pricing, taxes, subsidies, and quotas can help address water waste, biodiversity loss, resource depletion and environmental impacts (IFPRI 2016). Given that water used for agriculture represents 70 percent of all water withdrawals globally and, in some areas, agriculture accounts for about 80–90 percent of water use, the sector plays a crucial role in addressing water scarcity issues (FAO 2017). Likewise, enhancing the resource efficiency and environmental sustainability of food systems will contribute to climate change mitigation. For instance,

using alternatives to the flooding of rice paddies can reduce methane emissions by 45 percent, and adopting more efficient practices for livestock, such as zero tillage and integrated soil fertility management, can reduce emissions from the livestock sector by 40 percent (UNDP and UNRISD 2017).

Enhancing Climate Adaptation and Resilience to Environmental Shocks

In addition to reducing the environmental impacts of agriculture, the development of sustainable agricultural systems should also be designed to increase farmers' productivity and help farmers them adapt to climate change (IFPRI 2016). In particular, enhancing the agriculture sector's resilience and adaptive capacities by "climate-proofing" the farming systems of smallholder farmers and the rural poor will not only help communities, enterprises and institutions to adapt to the effects of climate change, mitigate and build resilience to climate change but also address the disproportionate negative impacts of climate change on the most vulnerable groups (FAO 2017). Without such investments, environmental factors, including climate change and land degradation, intensify inequalities, because those most affected by them are the impoverished and marginalized people (UNDP and UNRISD 2017). These inequalities exist across groups and across countries. Between 2010 and 2013, for instance, people living in least-developed countries were five times more likely to die from climate-related disasters than people living elsewhere (Pedrajas and Choritz, 2016).

Within countries, smallholder farmers, the majority of whom are poor, do not have access to the assets and services that could help them better handle unforeseen challenges. As a result, the poorest people who depend on agriculture for their livelihoods are more vulnerable to the vagaries of weather and climate-related shocks (IFPRI, 2016). Within this group, women are disproportionately affected by the impacts of climate change and environmental degradation, having more restricted access to resources and weak adaptive capacity, rendering them even more vulnerable and forcing them to engage in livelihood practices that can further damage the environment (Pedrajas and Choritz 2016).

In response to these factors, the development of sustainable agricultural production systems must prioritize interventions that not only mitigate climate impacts but also create the enabling environment for

the most vulnerable groups to strengthen their capacity for adaptation, while also improving soil and land quality, which will buttress the quest for resilience and adaptive capacity to climate change and environmental risks (UNDP and UNRISD 2017). Climate-smart agriculture, embodying a range of sustainable farming systems and practices, including conservation agriculture, agroforestry and integrated crop-livestock-energy systems, has the capacity to produce a triple-win by simultaneously improving agricultural productivity and helping farmers to adapt to the effects of climate change and mitigate agriculture's contribution to climate change (FAO 2017; IFPRI 2016).

Shifting to the widespread adoption of these practices will depend on a combination of community-level innovations, national interventions, and international support. This includes the promotion of innovation that generates growth which is based on increased efficiency as opposed to higher technological inputs. This type of growth is critical for achieving productive, efficient and sustainable agriculture especially in countries with a lot of smallholders (FAO 2017). Nevertheless, the scaled-up support from national governments and the international community will be critical for advancing sustainable practices to enhance resilience, particularly through investments in research and development and technological innovation (World Bank 2008). Already, technologies identified through such research, many of which can be used by smallholders, have helped safeguard livelihoods from the negative effects of climate change, while contributing to environmental sustainability; such technologies include zero-till farming, new crop varieties (such as C4 rice), and new agroforestry systems in which farmers grow trees and shrubs on their land (IFPRI 2016).

Incentivizing the use of climate-smart practices and technologies by smallholders could also be achieved through innovative financing and institutional support, such as conditional social safety net activities and the use of index-based insurance mechanisms. Continued efforts to reinforce climate adaptation and mitigation through the agriculture sector, including through the realignment of policies and incentive structures that run counter to sustainable agriculture, will be particularly decisive for the achievement of food and nutrition security and poverty eradication, along with the SDGs for environmental protection (FAO 2017).

Social Pathways

By contributing to a rise in inclusive and equitable growth, climate mitigation, and healthier environments, investments in sustainable agriculture and agrifood systems inevitably create space for individuals and communities to achieve greater access to the fundamental building blocks of human well-being, such as better health and educational opportunities due to improved incomes, reduced inequalities, and greater resilience to economic and climate shocks. Because these economic and environmental pathways support the social dimensions of human development, which, in themselves, contribute to the establishment of more peaceful and prosperous societies and enable individuals to achieve their full potential, it is likewise possible to observe the multidimensional role of agriculture within Agenda 2030 through its direct links to some of the underlying social drivers of development. Reinforcing the indivisibility of the entire framework, these social impacts constitute both outcomes of and inputs to the achievement of the economic- and environment-focused SDGs. While a range of potential socially focused pathways exist, two pathways in particular underline the role for agricultural production and food systems in accelerating progress across all the SDGs: the link between agrifood systems and nutrition and the opportunity to reduce gender inequalities through the empowerment of female smallholders.

Nutrition-Sensitive Agriculture and Food Systems

Good nutrition is fundamental for achieving many of the SDGs. While for young children it is critical for optimal brain functioning and the development of immune systems, its implications run across the life cycle and across generations, as it increases the chances of having a healthy and productive workforce in the future (IFPRI 2014). When considered in aggregate, these individual-level impacts have manifold consequences for countries' development agendas at large, playing a decisive role in the capacity for people to expand opportunities in life, including for employment.

A widening body of evidence has revealed the massive social and economic costs of malnutrition. The Cost of Hunger studies, for instance, demonstrate how failure to prevent undernutrition in children under the age of five directly results in “additional cases of morbidities, mortalities, school repetitions,

school dropouts and reduced physical capacity” in many countries, culminating in a quantifiable and often notable loss to national economies, sometimes as high as 16 percent of a country’s GDP (AUC and NEPAD 2017).

Despite receiving greater attention in national policies and international partnerships following on the momentum of such evidence, malnutrition remains one of the most urgent and central development challenges for all countries. Although the prevalence of undernutrition is declining through these heightened efforts, the pace remains inadequately slow. It is projected that with current trends, by 2025, there will still be 130 million stunted children, which is about 30 million above the global target (HLPF 2017a). Even more concerning, most countries must now grapple with a triple burden of malnutrition—that is, the coexistence of undernutrition, obesity, and micronutrient deficiencies within the same populations and sometimes even within the same families. Influenced by trends in urbanization, rising incomes, and shifting dietary habits, obesity continues to be on the rise, having nearly doubled in adults between 1980 and 2014 (HLPF 2017a).

Current agriculture policy still primarily focuses on production and trade rather than on biodiversity and nutritious quality, contributing to the fact that there are now as many overweight people as undernourished people in the world today (Giovannucci et al. 2012). The complexity of these challenges requires a multifaceted view of hunger and nourishment that considers all dimensions of food security, ensuring that access, utilization, and stability assume equal attention, along with investments in increased production. We know that food availability does not ensure food security. Although there is enough food for everyone on the planet, around 815 million people continue to suffer from hunger. Similarly, while food security may serve as a precondition for good nutrition, it does not guarantee it, as evident in the fact that the two billion people plagued by “hidden hunger,” or micronutrient deficiencies, is more than twice than those experiencing hunger.

As the primary source of food for all people, agriculture has an evident role to play in reducing malnutrition. Given that a substantial proportion of malnourished people reside in rural areas, policies that shape agriculture and food systems have a particularly potent effect on the rural poor, in terms of food

availability, accessibility, affordability and utilization by removing constraints on their food choices and nutritional intake (Global Panel on Agriculture and Food Systems for Nutrition 2014). Policies that focus only on supplying high-energy staple foods, for instance, which can lower their prices in relation to vegetables, pulses, fruit, and animal-based foods, most strongly affect those in rural areas with limited income and alternative food sources to achieve healthy diets (Global Panel on Agriculture and Food Systems for Nutrition 2014). As energy-rich, low-nutrient foods become more affordable to the poor worldwide, many developing countries are now dealing with empty calories while obesity is on the rise in many developed economies (ICSU 2017; FAO 2017).

At the same time, the role of agrifood systems in shaping dietary behaviors and the availability and access to nutritious foods is not confined to rural contexts. Already home to hundreds of millions of poor and hungry people, cities are increasingly seeing issues of overweight and obesity as trends of rapid urbanization alter not only consumer preferences but also the nutrient content of diets (IFPRI 2017). As more people continue to move to cities, the need for agrifood policies to coherently respond to malnutrition in its many manifestations in both rural and urban settings will be that much more acute. Strengthened rural–urban linkages through food value chains, for instance, can significantly alter the price of healthy foods, which can have far-reaching impacts on the diets of the urban poor, helping to expand their access to affordable healthy food options (IFPRI 2017).

On the production end of the food system, nutrition-sensitive agriculture, as integrated within broader multisectoral efforts and without sacrificing resilience or sustainability, represents a fundamental vehicle for reversing some of the trends that drive food choices for both rural and urban populations and that affect the intake of high-quality, nutritious diets to prevent malnutrition (Webb 2014). Much of this potential remains untapped, however, as agriculture policies continue to be largely defined in terms of more production as opposed to human nutrition (Giovannucci et al. 2012). Thus, these policies fail to use the multiple windows of opportunity through which food and agricultural pathways can play a greater role in promoting food quality and in creating the enabling environment for better nutrition (Webb 2014). Considered within the wider context of the food value chain, which encompasses all stages and actors

involved from production to consumption, the food system embodies several mutually reinforcing entry points for enhancing nutritional outcomes (IFPRI 2016).

At the base of the chain, policies can help support the production of more nutritious crops, such as through contractual relationships between smallholders and buyers, which can, in turn, stabilize prices and production. Likewise, financial incentives and market development that encourage smallholder farmers to produce indigenous, biodiverse foods would achieve multiple development objectives by boosting the livelihoods of vulnerable and marginalized smallholders, while sustainably increasing the supply of diverse, nutrient-rich foods (IFPRI 2016). Similar multifaceted impacts can be achieved along other points in the value chain, such as through the use of temperature-controlled supply chains to reduce food safety issues that could otherwise lead to food-borne diseases that weaken immune systems and nutrient intake. Overall, by helping to eradicate all forms of malnutrition, nutrition-sensitive agriculture and food systems will result in well-nourished, healthy individuals capable of reaching their full potential and contributing to inclusive and sustainable growth (Graziano da Silva 2016).

Empowering Women

Like good nutrition, gender equality acts as an enabler and an accelerator for all SDGs through its fundamental role in human development. It is impossible to achieve sustainable development as long as one half of humanity continues to be denied its full human rights and opportunities (Benson-Wahlen 2016; United Nations 2015). Yet, as complex barriers entrenched in the social, cultural, political, and economic landscapes of societies, gender inequalities continue to represent the most pervasive form of inequality limiting development across all countries. In terms of its capacity to directly tackle several dimensions of gender inequality for large segments of the population, a focus on agrifood systems—and on rural development, more broadly—can yield significant outcomes for women’s empowerment and, in turn, for the reduction of hunger, malnutrition, poverty, and environmental risks.

One of the most potent opportunities for achieving this impact rests with closing the gender gap in agriculture production by removing the policy and resource constraints faced by female smallholders.

The constraints faced by smallholder farmers, regardless of gender, already pose significant challenges for development. Although smallholders make up more than 90 percent of the world's 570 million farms and produce most of the world's food, they also make up the majority of the world's poor and hungry, with 84 percent of them operating on fewer than 2 hectares of land (FAO 2016a). Financial incentives and resources tend to favor large landholdings, despite smallholders supplying 40 percent of the world's food (Sekhran 2017). Within this disadvantaged group, women disproportionately bear the social and economic consequences of these barriers because of gender inequalities that influence the allocation of power and resources and, subsequently, the means of escaping poverty. Women are responsible for 60–80 percent of food production, yet they have less access than men to critical agricultural resources, such as land, credit, and modern technologies. In addition, they have less decision-making power, fewer opportunities for market integration, and decreased access to services that influence productivity and income-generating opportunities (Irish Aid 2017). Consequently, rural women tend to fare worse than rural men while women generally tend to fare worse than men on average on every gender and development indicator, including a higher risk of undernutrition and undernourishment. It is not surprising that approximately 60 percent of people living in hunger are female (FAO 2017).

While the importance of empowering smallholder farmers is clear for rural development, food production, and their impacts on all other areas of development, targeting this support to women would yield the greatest multiplier effects for sustainable development. Entry points for empowering rural women in agriculture include shifting the policy environment and legal frameworks to remove barriers to women owning land and other resources and increasing investments in rural infrastructure and labor-saving technologies that can help address the additional time burdens that women face through unpaid care and household work (UNDP 2016a; UNDP and UNRISD 2017). In particular, developing countries' statutory and customary laws greatly restrict the enabling environment for women farmers, including by barring their ownership to land, as seen in the fact that only 10–20 percent of landholders in developing countries are women (UNDP 2016a). The policy frameworks and surrounding social norms likewise affect the distribution of power and the potential for women to gain economic independence, with as many as

30 percent of women not allowed to make economic decisions in their own households (IFAD 2016; UNDP 2016a). More inclusive policies would thus go a long way in empowering rural women to help make decisions that affects their lives, as well as granting them access to the fundamental resources needed to achieve the same level of production and economic opportunities in agriculture as men (IFAD 2016).

Complementary to addressing the policy barriers, interventions that improve women's access to food value chains and markets would increase their commercial opportunities (IFAD 2016). This includes providing women with the tools, knowledge, and technical skills to handle financial and risk management, helping them connect with high-value markets, increasing their resilience to shocks, fostering their leadership in farmers' organizations, and extending their access to investment opportunities, extension services and capacity development (IFPRI 2016). The outcomes of these policy measures and other interventions to empower women smallholders would not only have a direct impact on social and economic conditions at the individual and household levels but would also have a trickle-down effect on substantial portions of both rural and urban populations through the boost to agricultural productivity and inclusive economic growth.

Estimates from available country data suggest that eliminating gender gap in agriculture can possibly increase national production by 2.5 to 4 percent (HLPF 2017b). Such growth can also lead to more environmentally sustainable outcomes, particularly when efforts are made to include women in managing natural resources (IFPRI 2016). Likewise, the implications for the achievement of SDG 2 would be significant, as it is estimated that the number of hungry people in the world could be reduced by up to 150 million if both women and men have the same resources (Sekhran 2017). More specifically, given the childcare role often assumed by women, there is a greater likelihood improvements in household income, coupled with greater decision-making power for women, would lead to improved nutritional outcomes among children; overall, it could also lead to declines in rates of undernourishment by 12–17 percent (HLPF 2017b).

4. CONCLUSION: BRINGING ALL THE PIECES TOGETHER

As much as the pathways through which agricultural systems influence outcomes in primarily social-, economic-, or environmental-focused development goals helps shed light on the sector's multidimensional potential to contribute to Agenda 2030, they likewise reveal how no single dimension exists in isolation from the other two. Equally so, they show that failure to invest in one pathway can undermine efforts in another, just as progress in one can multiply the achievements in the other two. For example, as illustrated in this paper, investments in increased agricultural productivity can result in significant economic growth. However, without simultaneous investments in technologies that conserve natural resources or the promotion of sustainable methods of production, the economic gains produced by the sector will be short-lived, in that the resulting land degradation, water scarcities, and increased occurrence of climatic shocks will undermine livelihoods and constrain capacities for further agricultural production to meet the rising needs of the global population.

It is in this embodiment of interconnectivity that agrifood systems can facilitate the achievement of SDG 17, which renders the means of implementation as important as the end outcomes. Most notably, agrifood systems can help operationalize the targets aimed at tackling systemic issues, including by acting as a catalyst for enhancing policy and institutional coherence, as well as serving a strong platform for fostering multistakeholder partnerships. For example, efforts to strengthen the inclusivity and sustainability of food systems across all points of the value chain naturally implicate both public and private actors, from smallholder farmers to the institutions that shape the policy environment and production systems. A multitude of opportunities for furthering private-public partnerships also exist in the need for the continued mobilization of research and development that can advance the sustainability agenda of agriculture.

At the level of governance, agriculture can help strengthen integration by placing a spotlight on the intrinsic links between rural and urban issues, as well as on the links between sectors. For instance, just as cities are becoming catalysts for agriculture, agriculture must adapt to supply this growing urban

population, especially as two-thirds of the world's population is projected to live in cities by 2050, placing additional demands on local food supplies (FAO 2017). This dynamic, not to mention the potential costs for governments should they fail to address these trends, provides strong incentive for ensuring that policies respond to both urban and rural development challenges holistically.

At the same time, the need for national policies to respond to locally specific issues that shape agriculture can help facilitate greater vertical coherence or integration across various levels of government, including through platforms to raise the voices of citizens and civil society in policy planning. Agriculture can thus serve as an important vehicle for delivering locally led development. Agriculture can help strengthen policy and institutional coherence by way of necessity: only policies that consider the multiple ways by which progress in agriculture affects (and is affected by) its relationship with rural and urban populations, inclusive economic growth, the empowerment of marginalized groups, gender equality, education, governance and legal systems, and the establishment of peaceful societies will yield effective and sustainable outcomes.

In encompassing a range of distinct yet interrelated roles, not only for food production systems but also for such issues as social organization, land-access issues, the interlinkage of rural and peri-urban environments, cultural identities, and local and indigenous knowledge, agriculture provides a channel for bridging a range of interconnected facets of development (FAO and IFAD 2013). It is thus not in the sector-specific development outcomes that agriculture has its greatest role in supporting the achievement of Agenda 2030 but, rather, in its capacity to connect the dots, optimize synergies, and serve as a multiplier across most of the SDGs. When approached holistically, agriculture, by its nature, can help elevate the integration imperative called for by the SDGs.

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